Week 12

Course Level Objectives:

- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine whether a study calls for one-proportion or difference-in-proportions inference. (CLO 5) **2.** Check the conditions for inference about a difference in proportions. (CLO 5) **3.** Construct a confidence interval for $p_1 - p_2$. (CLO 6) **4.** Interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CLO 6) **5.** Compute the pooled proportion. (CLO 8) **6.** Perform a two-proportion z test from summary data or a two-way table. (CLO 8)	e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT) Interactive Lecture: 6.3 Inference for a Difference in Proportions Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	Participation: Difference in Proportions – two-proportion randomization (18/30 vs. 12/30); compare simulation with the formula Your Turn: Treatment vs Placebo (two-proportion z test from a two-way table)	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM) Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
7. Explain why inference for a mean uses the t distribution with df = n − 1. (CLO 5) **8.** Check the conditions for t procedures. (CLO 5) **9.** Construct and interpret a t confidence interval for a population mean. (CLO 6) **10.** Perform a one-sample t test and state the conclusion in context. (CLO 8)	e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT) Interactive Lecture: 6.2 Inference for a Mean Slides: 6.2 Inference for a Mean StatKey – Theoretical Distributions (t)	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups Your Turn: Testing a Mean (juice bottles underfilled?)	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM) Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 6.1–6.3 (all objectives) Test 3 (Fri Nov 20 – Sun Nov 22, proctored) – covers 4.4–6.5 (all objectives) Test 3 Study Guide (all objectives) Practice Test 3 (optional; due Thu Nov 19) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus